

Milk is not just food but most likely a genetic transfection system activating mTORC1 signaling for postnatal growth



Milk has been recognized to represent a functionally active nutrient system promoting neonatal growth of mammals. Cell growth is regulated by the nutrient-sensitive kinase mechanistic target of rapamycin complex 1 (mTORC1). There is still a lack of information on the mechanisms of mTORC1 up-regulation by milk consumption. This review presents milk as a materno-neonatal relay system functioning by transfer of preferential amino acids, which increase plasma levels of glucose-dependent insulintropic polypeptide (GIP), glucagon-like peptide-1 (GLP-1), insulin, growth hormone (GH) and insulin-like growth factor-1 (IGF-1) for mTORC1 activation. Importantly, milk exosomes, which regularly contain microRNA-21, most likely represent a genetic transfection system enhancing mTORC1-driven metabolic processes. Whereas human breast milk is the ideal food for infants allowing appropriate postnatal growth and species-specific metabolic programming, persistent high milk signaling during adolescence and adulthood by continued cow's milk consumption may promote mTORC1-driven diseases of civilization.